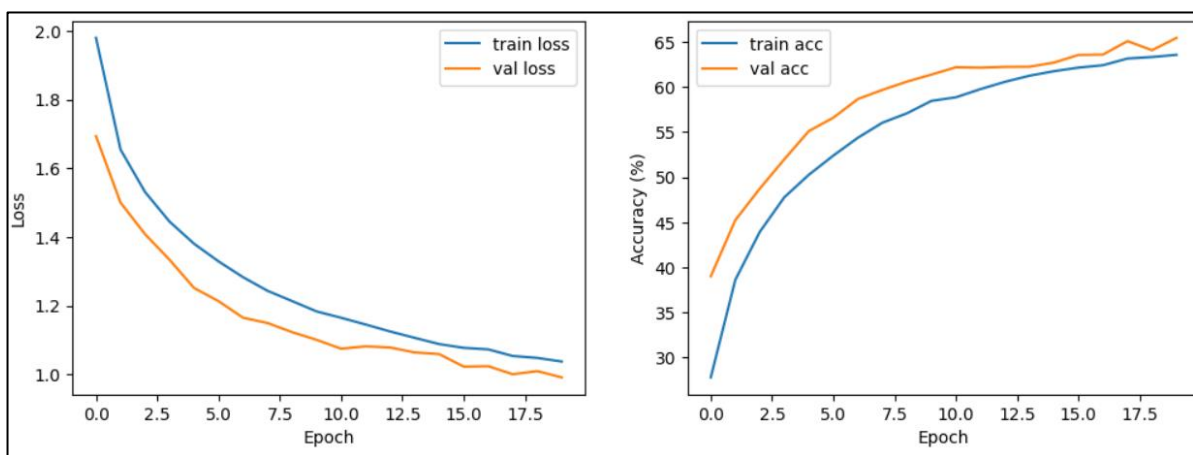


During the lab, the most challenging aspect was understanding PyTorch's coding structure and implementation workflow. AI assistance was useful in clarifying difficult concepts, while the official PyTorch documentation, particularly the guide on training a classifier, provided valuable support.

A basic CNN model was implemented using

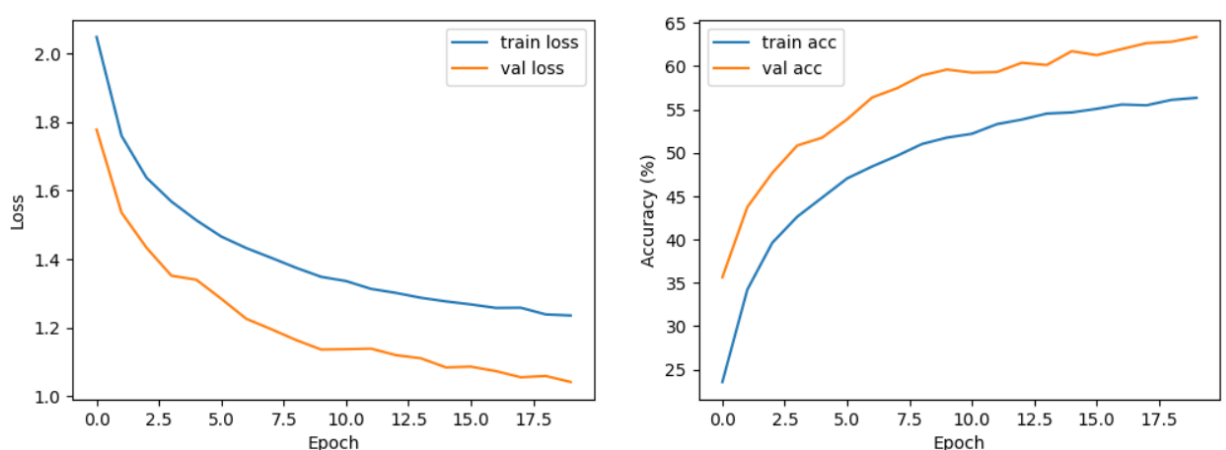
1. **Batch Normalization**
2. **a learning rate of 0.001**
3. **Dropout Layer**
4. **L2 regularization.**

The resulting performance is shown in the following plot.



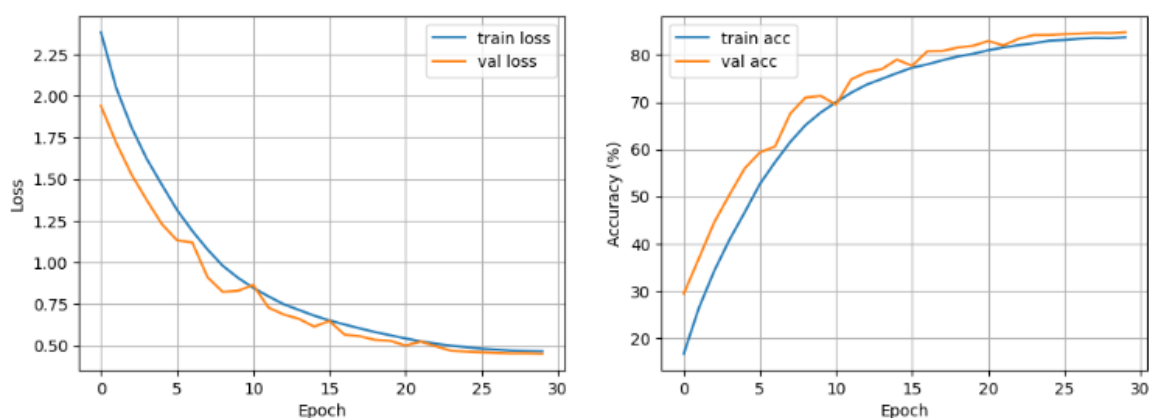
Maximum of 65 percent of accuracy was achieved but no over fitting which is a good sign.

Also by adding data augmentation to the input training images we got the following plot:



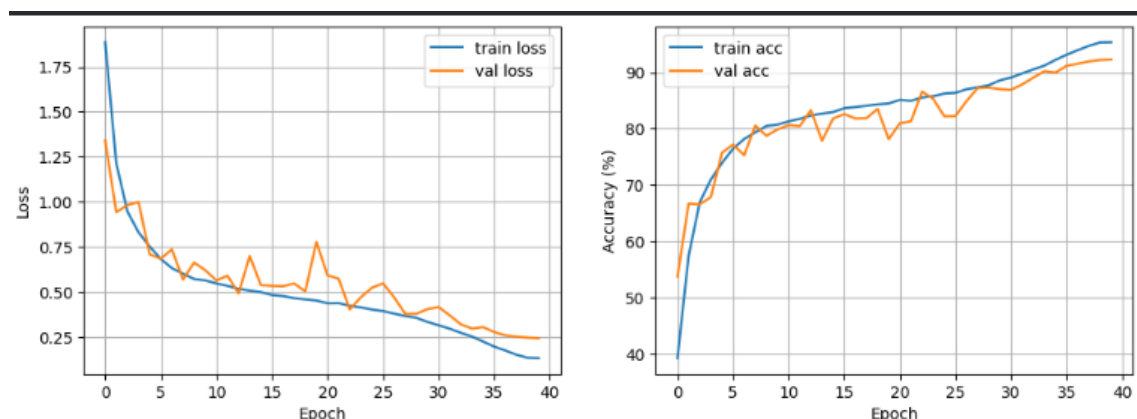
We got a much lower accuracy than before more trials are needed.

This means the model need a little bit of depth to better understand the features much better, so in the next one we add more convolution layers in sections of the model and increase the no of epochs to 30.



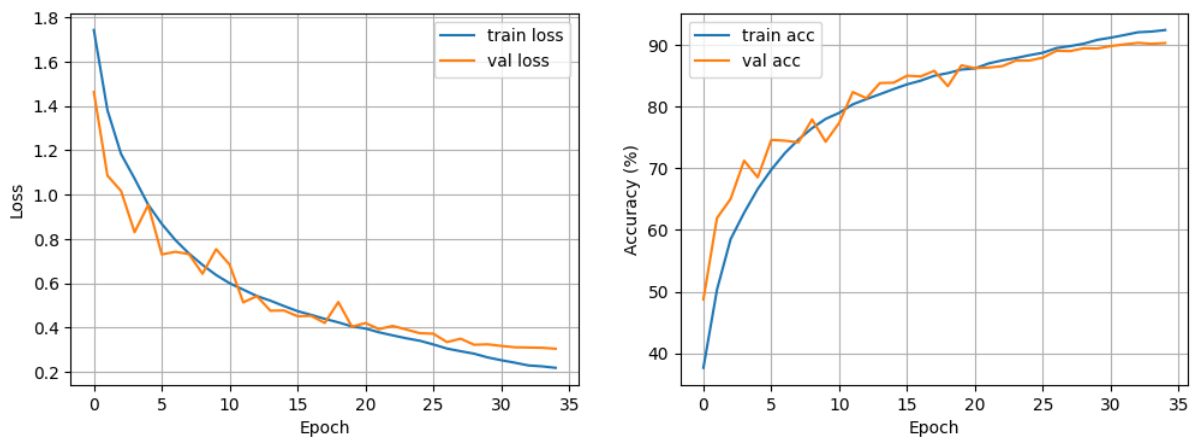
By applying the previously discussed techniques and increasing the number of convolutional layers, the model achieved an accuracy of approximately 81% without significant overfitting. However, the training curves remained somewhat jittery, indicating that the learning rate could be reduced and the number of training epochs increased for more stable convergence.

In the next experiment, the optimizer was changed from SGD to Adam, with a learning rate of 2×10^{-4} . The resulting training and validation curves are shown below.



By changing the optimizer and increasing the number of epochs, the model achieved a higher accuracy. However, the training curve became more jittery, and the validation accuracy remained slightly lower than the training accuracy, indicating a small degree of overfitting.

Therefore, we retained the SGD optimizer, modified the convolutional layer architecture, and slightly increased the learning rate. The resulting training and validation curves are shown below.



Initially, validation accuracy is higher than training accuracy because dropout and data augmentation are applied during training, making the training data more difficult. Validation is performed in evaluation mode without these effects.

We see that the jagged curves is mainly do to the model having fixed learning rate which caused the model to learn with pits and bums but as the training progresses the cures goes smooth which means the model learned the features correctly but there is only so much we can achieve for this type of plan CNN model